Data handling & Visualization | Activity - 7 | 22 to [illegible] | Y. Veeranadan Reddy, 192305102, Slot-C

1) Correlation Matrix:-

* No. of observations : n = 14
* No. of variables : 7
* No. of distinct variable pairs : $\frac{7(7-1)}{2} = 21$.

Variable	temp	Ice cream	rescues	Pool dev	complaints	Sunscreen
temp	1.00	0.80	0.86	-0.17	0.08	-0.49
Ice cream	0.80	1.00	0.85	-0.30	-0.07	-0.25
rescues	0.86	0.85	1.00	-0.32	0.14	-0.28
Pool dev	-0.17	-0.30	-0.32	1.00	0.03	-0.12
Complaints	0.08	-0.07	0.14	0.03	1.00	-0.15
Sunscreen	-0.49	-0.25	-0.28	-0.12	-0.15	1.00

2) * Strongest Positive & Negative Correlations :-

* Strongest +ve pair :- temp & rescues, $r = 0.86$ This is a strong +ve correlation.
* Strongest -ve pair :- temp & Sunscreen, $r = -0.49$.
This is a moderate -ve correlation

3) * The Deceptive Zero :-

* Correlation b/w Complaints & pool-dev : $r = 0.03$
* The linear correlation is approximately zero, suggesting almost no straight-line relationship
* quadratic fit : Complaints ~ pool_dev + I(pool_dev^2)
* R^2 of quadratic fit : approximately 0.93.

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complaints increase when pool temperature deviates from 27°C in either direction. The quadratic model explains approximately 93% of variation.

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4. The Outlier Alibi:—

* Wrong record : Row 3
* In Row 3, lifeguards = 16 and Sunscreen = 180.
* Correlation using all 14 rows : $r = 0.60$
* Correlation after removing row : $r = 0.15$.

Conclusion: Row 3 is an influential outlier that substantially inflates the correlation b/w lifeguards and sunscreen sales. without that row.

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5. The prime Suspect:—

Correlation values:

* r(ice cream, rescues) = 0.85,
* r(temp, icecream) = 0.80,
* r(temp, rescues) = 0.86.

Partial correlation: After controlling for temperature by correlating the residuals of both variables, the partial correlation is approximately.

$$r_{\text{ice cream, rescues | temp}} \approx 0.56$$

Verdict:- Icecream - sales and rescue calls are strongly correlated, but temperature is shared variable. The remaining the correlation is not explained entirely by temperature.

Study design to test causation:- Conduct a randomized controlled study in which comparable beach locations receive randomly assigned ice-cream promotions, then compare rescue rate while controlling for temperature, visitor rols, & lifeguard staffing..